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COMP 4421 (Spring 2025)

Assignment 1

Department of Computer Science and Engineering
The Hong Kong University of Science and Technology

Total = 100 marks

Due: 11:55 pm, March 26, 2026

Assignments must be submitted via CANVAS

Late Policy: Late assignments incur a 10%-score penalty; only one day late is allowed.

Programming Assignment

In this assignment, you should complete the Python functions in the given Jupyter Notebook. Download `programming1.ipynb` and related files from CANVAS.

1. Your submitted Jupyter Notebook must be named `programming1.ipynb`.
2. Runtime requirement: your notebook must finish within 5 minutes.
3. Compress all files into one file named
[your 8-digit student ID]_assign1.rar (or zip)
Example: 09654321_assign1.rar
4. If multiple submissions are made before the deadline, the latest version will be graded.
5. Plagiarism is taken seriously. You may discuss ideas but **must not share code or pseudocode**.
6. Marks will be deducted for requirement violations.

Grading criteria

- Correctness and efficiency of algorithms.
- Adherence to assignment. 20% penalty for any forbidden extra library imports.

Written Assignment

Submit your written assignment as a PDF named

[your 8-digit student ID]_w-assign1.pdf

Example: 09654321_w-assign1.pdf

Include this file in the same compressed submission.

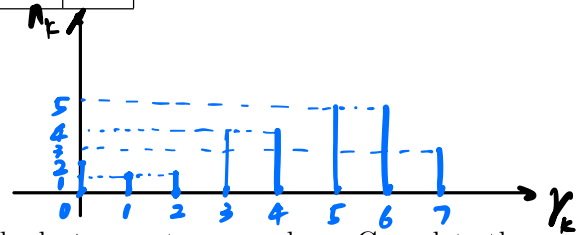
Question W1: Histogram Equalization

Given an image I of size 5×5 pixels with intensity levels ranging from 0 to 7 (8 grey levels). Round your answers to **2 decimal places**, unless otherwise specified.

Input image

6	7	6	5	6
3	0	3	6	7
4	1	2	5	7
0	3	4	4	5
5	4	5	3	6

- (a) Sketch the histogram for the image I .



- (1) Perform histogram equalization based on the lecture notes procedure. Complete the intermediate table and round the final mapped intensity to the nearest integer.

Intensity Level k	0	1	2	3	4	5	6	7
Pixel Count n_k	2	1	1	4	4	5	5	3
Probability $p(r_k) = n_k/25$	$\frac{2}{25}$	$\frac{1}{25}$	$\frac{1}{25}$	$\frac{4}{25}$	$\frac{4}{25}$	$\frac{5}{25}$	$\frac{5}{25}$	$\frac{3}{25}$
Cumulative Probability $\sum_{j=0}^k p(r_j)$	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{4}{25}$	$\frac{8}{25}$	$\frac{12}{25}$	$\frac{17}{25}$	$\frac{22}{25}$	1
Mapping $s_k = 7 \sum_{j=0}^k p(r_j)$	0.56	0.84	1.12	2.24	3.36	4.76	6.16	7
Final Intensity (Rounded s_k)	1	1	1	2	3	5	6	7

Output Image

6	7	6	5	6
2	1	2	6	7
3	1	1	5	7
1	2	3	3	5
5	3	5	2	6

- (c) Histogram specification adjusts the histogram to match a specified probability distribution. Given the following distribution, perform histogram specification and fill in the output image in integer intensities.

Specified Distribution CDF

Target Intensity Level j	0	1	2	3	4	5	6	7
Target Probability $P_z(z_j)$	0.25	0.2	0.1	0.1	0.1	0.1	0.1	0.05
Target Cumulative $\sum_{i=0}^j P_z(z_i)$	0.25	0.45	0.55	0.65	0.75	0.85	0.95	1

Intensity Mapping

1.75 3.15 3.85 4.55 5.25 5.95 6.65 7
2 3 4 5 5 6 7 7

Input Intensity r_k	0	1	2	3	4	5	6	7
Equalized Value $T(r_k)$	1	1	1	2	3	5	6	7
Matched Output Intensity z_j	0	0	0	0	1	3	5	7

Output Image

5	7	5	3	5
0	0	0	5	7
1	0	0	3	7
0	0	1	1	3
3	1	3	0	5

Question W2: Sobel Filtering

Given the following 7×7 grayscale image I . Use the corresponding 3×3 Sobel operators. Round your answers to **2 decimal places**.

0	0	0	0	0	0	0
1	0	0	0	0	0	0
1	1	0	0	0	0	0
1	1	1	<u>1</u>	0	0	0
1	1	1	1	1	0	0
1	1	1	1	1	1	0
1	1	1	1	1	1	1

- (a) Consider the 3×3 neighborhood centered at the underlined pixel 1. Compute the Sobel horizontal gradients G_x for each pixel in this 3×3 region.

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

-2	-1	-1
-1	-2	-3
0	-1	-3

- (b) Compute the Sobel vertical gradients G_y for the same 3×3 region.

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

4	3	1
3	4	3
0	1	3

- (c) Compute the **gradient magnitudes** $G = \sqrt{G_x^2 + G_y^2}$ for the same 3×3 region.

4.47	3.16	1.41
3.16	4.47	4.24
0	1.41	4.24

Question W3: Bilateral Filtering

Perform bilateral filtering on the input image. You do not need to fill in * cells because they correspond to zero-padding.

- Spatial sigma = 1
- Range sigma = 5
- Kernel size = 3

Round values to **2 decimal places**.

Input image

0	0	0	0	0	0
0	1	1	0	1	0
0	1	1	0	1	0
0	1	1	0	1	0
0	0	0	0	0	0

Output image

*	*	*	*	*	*
*	0.53	0.53	0.39	0.33	*
*	0.73	0.73	0.54	0.46	*
*	0.53	0.53	0.39	0.33	*
*	*	*	*	*	*

$$d = \begin{bmatrix} \sqrt{2} & 1 & \sqrt{2} \\ 1 & 0 & 1 \\ \sqrt{2} & 1 & \sqrt{2} \end{bmatrix}$$

$$f_s = e^{-\frac{d^2}{2\sigma_s^2}} \quad \sigma_s = 1$$

$$f_r = e^{-\frac{(I(q) - I(p))^2}{2\sigma_r^2}} \quad \sigma_r = 5$$

$$w(q) = f_s(q) \times f_r(q)$$

$$I'(p) = \frac{\sum I(q) \cdot w(q)}{\sum w(q)}$$